C-HI 5 Myr XRAiSE Grid Smoke

Date: 2026-05-10

Summary

This report records the first XRAiSE grid-thermal X-ray smoke product for the particles-enabled high-power cluster case, `C-HI`, using the same target as the truth, bridge observable, and cavity-protocol smokes:

- grid output: `data.0008.flt.h5`
- grid timestep: `Timestep_8`
- physical time: `4.9994 Myr`
- view: `0 deg`, projected along the simulation `y` axis
- photon energy: `1 keV`
- redshift: `0.05`

The extraction was run on a `kunanyi` compute node as PBS job `690624.kunanyi-ohpc` and completed successfully. This is the first smoke that uses the supplied XRAiSE Python source plus the supplied APEC and Gaunt correction tables. It supersedes the earlier C bridge map for the grid-thermal X-ray component, but it does not yet include the heavier particle-loss kernel.

What Changed From The Bridge Smoke

The earlier observable bridge smoke ported Georgia's XRAiSE bremsstrahlung expression before the APEC/Gaunt correction stage. This run uses:

- `xraise.py`, copied from Patrick's supplied `env/graile-all.zip`
- `apec_corrections.txt`, copied from `apec_corrections(1).txt`
- `gaunt_lut_2d.npz`
- a small streaming HDF5 adapter that scans `data.0008.flt.h5` in `y` chunks and projects along the line of sight

The run uses the TPAC `Anaconda3/2024.02-1` module. That module already provides the Python stack needed for this grid-only XRAiSE step: `numpy`, `h5py`, `scipy`, `astropy`, `numba`, `matplotlib`, `yaml`, `pandas`, and `IPython`.

Kunanyi Job

- job id: `690624.kunanyi-ohpc`
- job name: `c_hi_xrgrid5`
- node: `n001`
- resources: `select=1:ncpus=1:mpiprocs=1`
- walltime requested: `00:45:00`
- extraction window: `2026-05-10T09:03:17Z` to `2026-05-10T09:05:10Z`
- extraction result: no stderr output, archive written successfully

Compact archive:

```
remote: /u/bwbrook/codex_kunanyi/c_hi_xraise_grid_smoke_2026-05-10.tar.gz
local:  build/science/kunanyi_stage_2026-05-08/c_hi_xraise_grid_smoke_690624_compact_2026-05-
sha256: 85f36265a1563483a582bb904d69094fffce2811f45493f7c38bf53545c6ca23
```

Scalar Smoke Metrics

Metric	Value
grid time	`4.99943481897884 Myr`
grid shape	`452 x 404 x 404`
raw pre-correction 1 keV max	`1.4382e6 mJy sr^-1`
XRAiSE-corrected 1 keV max	`2.8783e6 mJy sr^-1`
raw pre-correction map sum	`7.9961e8 mJy sr^-1 pixels`
XRAiSE-corrected map sum	`1.0937e9 mJy sr^-1 pixels`
correction-ratio range	`1.3273` to `2.4404`

The raw pre-correction value matches the earlier C bridge smoke, which is a useful cross-check that the Python XRAiSE path and the bridge extractor are using the same base thermal expression before table corrections.

Figure

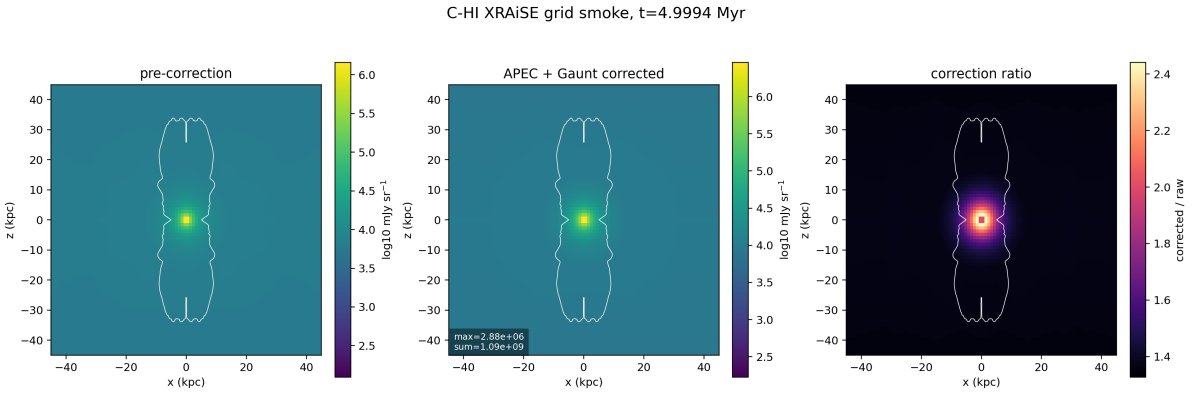


Figure 1: C-HI XRAiSE grid smoke maps

The white contour marks the line-of-sight maximum tracer boundary at `tracer = 5e-3`. The right panel shows the multiplicative correction ratio from the supplied APEC and Gaunt tables.

Products

The compact products are stored under `docs/science/assets/c_hi_xraise_grid_smoke_2026-05-10/`:

- `xraise_grid_smoke_summary.csv`
- `xraise_grid_smoke_metadata.json`
- `xray_1kev_raw_mjy_sr.tsv`
- `xray_1kev_xraise_corrected_mjy_sr.tsv`
- `xray_1kev_correction_ratio.tsv`
- `tracer_max_los.tsv`
- `x_kpc.tsv`
- `z_kpc.tsv`
- `c_hi_xraise_grid_smoke_maps.png`

The reusable scripts are:

- `scripts/science/xraise_grid_smoke.py`
- `scripts/science/plot_xraise_grid_smoke.py`
- `scripts/kunanyi/c_hi_xraise_grid_smoke.pbs`

Interpretation

This passes the XRAiSE grid-thermal smoke gate:

- the supplied XRAiSE source and correction tables can run on a `kunanyi` compute node using the site Anaconda module
- the same aligned `5 Myr`, `0 deg` target can now produce an APEC/Gaunt-corrected 1 keV X-ray map
- the pre-correction map reproduces the previous bridge value, anchoring the port against the earlier smoke
- the correction ratio is finite across the projected map and strongest near the bright central region

The remaining GRAiLE/XRAiSE upgrade is the particle-loss-aware component. The current supplied driver expects a preprocessed `*.particles.hdf5` file and a `plutokore` runtime; on `kunanyi`, those are still the next runtime gates.